Unit: 6 Unit 9: Data Statistical Analysis
Topic: Measures of Central Tendency

Name: ________________
Date: ___________

Advanced (Carolina Reaper)

Q1. What is the mean of the numbers $2, 4, 6, 8, 10$?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 10

Q2. A dataset has a median of 15 and a mode of 20. What can be said about the data?
(A) The data is symmetric
(B) The data is skewed to the left
(C) The data is skewed to the right
(D) The mean is 17.5
(E) The range is 5

Q3. What is the mode of the numbers $1, 2, 2, 3, 3, 3$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) No mode

Q4. A set of exam scores has a mean of 80 and a standard deviation of 10. What is the z-score of a student who scored 90?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q5. Which measure of central tendency is most affected by outliers?
(A) Mean
(B) Median
(C) Mode
(D) Range
(E) Variance

Q6. What is the median of the numbers $1, 3, 5, 7, 9$?
(A) 3
(B) 5
(C) 7
(D) 9
(E) 11

Q7. A dataset has a mean of 12 and a median of 10. What can be said about the data?
(A) The data is symmetric
(B) The data is skewed to the left
(C) The data is skewed to the right
(D) The mode is 12
(E) The range is 2

Q8. Which of the following is a characteristic of the mode?
(A) It is always unique
(B) It is always the same as the mean
(C) It is the most frequently occurring value
(D) It is sensitive to outliers
(E) It is only used for categorical data

Open Ended Questions (FRQ)

Q9. A dataset has the following values: $2, 4, 6, 8, 10, 12, 14$. Calculate the mean, median, and mode. Show your work and explain your answers.

Q10. Compare and contrast the mean, median, and mode. Explain when each measure is most appropriate to use and provide examples. Show your work and explain your answers.

Chef's Tips

- Emphasize the importance of showing work and explaining answers
- Encourage students to use real-world examples to illustrate their points
- Remind students to check their calculations carefully